MILITARY SPECIFICATION

VACUUM PUMP, HAND OPERATED FOR ARMY OIL ANALYSIS PROGRAM (AOAP)

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, Engineering Support Directorate, Tool and Equipment Engineering Division, Product Engineering Branch, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a hand operated vacuum pump used in drawing samples of lubricating oil, transmission-fluid, and hydraulic fluid from engines, transmissions, power train components and accessories for analysis, as required under the Army Oil Analysis Program.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

TB43-0211 - Army Oil Analysis Program, Users Guide, Non-aeronautical Equipment

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-A, Rock Island, IL 61299-7300 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Packaging Waterproof
PPP-B-566 - Boxes, Fiberboard, Wood-Cleated
PPP-B-591 - Boxes, Shipping, Fiberboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676 - Boxes, Set-up

MILITARY

MIL-M-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-M-7298 - Manuals Technical: Commercial Equipment
MIL-F-22191 - Barrier Materials, Transparent Flexible, Heat Sealable
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Artic

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables
MIL-STD-130 - Identification Marking of Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing, and Marking

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2911-82 - Standard Specification for Dimensions and Tolerances for Plastic Bottles

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish one or more complete hand operated vacuum pumps (with the bottle and tube) for first article inspection. The first article may be either a preproduction model(s) or an initial production item(s) which conforms to the requirements herein. In either case, the approved first article and the production article shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in the contract in accordance with the requirements herein. Only the vacuum pump is required to be delivered under this specification. The bottle and tube are required for testing purposes only and are not to be delivered.

3.2 Design. The vacuum pump shall be new, hand operated, piston type, and of the manufacturer's latest design meeting all of the requirements specified herein.

3.2.1 Maintainability. Item configuration and features will be designed to reduce maintenance to simple procedures requiring a minimum of maintenance skills. The pump shall be easily disassembled and reassembled for cleaning, without tools, and shall require no tools for operation. Instructions for disassebly, repair, and reassembly will be concise and clearly defined in an instruction and repair parts manual.

3.3 Construction. The vacuum pump shall be clean, new, and constructed of materials imprevious to engine lubricating oils, transmission and hydraulic fluids. All surfaces on the pump shall be free of burrs, sharp edges, and extraneous materials. The pump shall function and shall not be damaged when sampling oil and fluids of temperatures up to 25(3 degrees F. The pump shall be for use with oils of grades OEA of MIL-L-46167 (artic grade oil) and OE/HDO-15/40 of MIL-L-2104.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with the best commercial practices within the hand vacuum pump industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore then to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

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3.3.3 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances. All respective parts of the end item shall be interchangeable without requiring modification of the replaced part or its mating components. All parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in the repair parts manual. All replacement, repair and spare parts shall provide the same or better form, fit, function, and performance as the parts provided initially installed in the end item. When industry or Government standards are applicable, all replacement parts shall conform to the standards.

3.4 Performance and product characteristics.

3.4.1 Capacity. The vacuum pump shall be capable of siphoning oils of grades OEA to OE/HDO-15W/40 at temperatures up to 250 degrees F, to a height not less than 60 inches, and fill a 100 milliliter sample bottle, using 1/4 inch outside diameter plastic tubing as indicated in Technical Bulletin 43-02211. The pump shall operate in accordance with the requirements stated herein, with the ambient atmospheric temperatures varying anywhere within the range of 0 to 120 degrees F. The above function must be completed within three completed strokes of the piston, and within the time frame of one minute.

3.4.2 Proof vacuum. The vacuum pump shall be capable of producing and maintaining a vacuum of at least 25 inches of mercury for not less than 1 minute.

3.5 Details of components.

3.5.1 Pump assembly. The pump assembly shall be designed for hand operation by one person, and shall incorporate a check valve in the pump assembly to maintain vacuum in the pump head. The pump shall be designed so that the sample fluid does not contact any part of the vacuum pump but shall contact only the sampling tubing or sample bottle under normal operating conditions.

3.5.2 Seals. The pump piston seals shall be fabricated from materials compatible with the oils specified in 3.3.

3.5.3 Pump head. The pump head shall be the portion of the vacuum pump to which the sampling tubing and sampling bottle are attached. The pump head shall have an internal screw thread for attachment of a sample bottle. The thread dimensions shall conform to finish thread L43SP400 of ASTM D2911-82. The pump head shall also have a connector designed to permit a non-metallic tube of 1/4 inch outside diameter to be inserted thru the pump head, into the sampling bottle, and be capable of providing a seal around the entire circumference of the tubing. The connector shall not require tools for tightening. The pump head shall have a self closing vacuum release device. When the sample bottle is filled to the required level, the operation of the vacuum release device shall stop the fluid flow and allow the fluid in the tube to drain back into the crankcase, transmission or fluid reservoir. The pump head shall attach to the pump and shall be removable for cleaning. There shall be no deterioration of the function of the pump head due to heat or contact with the fluids being sampled.

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3.5.4 Product identification plate. The hand pump shall be marked for identification in accordance with MIL-STD-130 and shall contain the following information.

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

The product identification plate shall be fastened in a permanent manner and shall be corrosion resistant with permanent and legible characters in the English language.

3.5.5 Workmanship. The quality of workmanship imparted to the item and all of its components shall equal or exceed that typically provided to commercial products by domestic producers of the type of item addressed herein. The items presented for acceptance shall have been manufactured with skill and care; shall be uniform, neat, and clean; and shall be free of l irregularities and anomalies which degrade form, fit, function, performance or appearance.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1)

(b) Quality conformance inspection (see 4.2.2)

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4.2.1 First article inspection. When specified, first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.3, and the tests of 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection.

4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist-of (a) through (c) below. Failure of any unit to pass an examination or test shall be cause for rejection of the unit.

(a) Product examination (see 4.3)

(b) Vacuum test (see 4.4.2)

(c) Packaging inspection (see 4.5)

4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following.

Inspection or test	Level	AQL
Product examination	II	1.0
Vacuum test	II	1.0
Packaging inspection	S-4	4.0

4.3 Product examination. The vacuum pumps shall be visually and manually examined to determine conformance with the requirements of 3.2 through 3.3.3 and 3.5 through 3.5.5. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination. provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the vacuum pumps and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit or function of the end item (see 3.3.3), shall be cause for rejection.

4.4 Tests.

4.4.1 Performance test. The pump shall be operated to draw through the 1/4 inch OD plastic tubing supplied with the pump, oil from a container situated at least 60 inches below the pump. The-oil shall conform to grade OE/HD-15W/40 of MIL-L-2104, and shall be at a temperature of no less than 250 degrees F. When the sample bottle has been filled with not less than 100 milliliters of oil, the vacuum release device shall be operated to stop the flow of oil into the sample bottle and allow the remaining oil to drain from

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the tubing. Inability to draw at least 100 milliliters of oil up the tube and into the sample bottle in one minute or less with not more than three full strokes of the pump handle, or failure of the vacuum release device to operate as specified, shall be cause for rejection (see 3.4.1 and 3.5.3).

4.4.2 Vacuum test. Each pump shall be attached to a vacuum gage which is accurate within + .25 inch of mercury. The pump shall then be operated to produce a vacuum of at least 25 inches of mercury, and the vacuum retained for at least one minute without further operation of the pump. Failure of the pump to produce a vacuum of at least 25 inches of mercury and retain the vacuum for not less than one minute without further operation shall be cause for rejection (see 3.4.2).

4.5 Packaging inspection. The vacuum pump shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5 and in Part I - Section D of the contract.

5 PACKAGING

5.1 Packaging. Packaging shall be level A or Level C, as specified (see 6.2) .

5.1.1 Level A. Each pump shall be heat sealed within a snug-fitting film bag of 0.0015 inch gage material conforming to MIL-F-22191, Type I. Each pump, prepared as aforementinoed, shall be preserved and cushioned in a sealed water-resistant fiberboard container. No contact preservative is required. Each pump shall be packaged within a snug, holding, or set up box conforming to PPP-B-566 or PPP-B-676. Forty unit packages shall be packaged within a fiberboard container conforming to PPP-B-636, Type CF, also weather resistant, grade V3C, style RSC. Closure and sealing of the container shall be in accordance with the appendix to PPP-B-636. All seams and joints shall be sealed with tape conforming to PPP-T-60. Technical data shall be packed in accordance with method IC-1 of MIL-P-116 and placed inside the unit container.

5.1.2 Level C packaging. Level C packaging shall be in accordance with MIL-STD-1190.

5.2 Packing. Packing snail be level A, B or C as specified (see 6.2). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A. The pumps packaged as specified in 5.1.1, shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-636, or PPP-B-640 Class 2. Closure and strapping of the containers shall be in accordance with the applicable containers specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitation of the applicable container specification.

5.2.2 Level B. The pumps, packaged as specified in 5.1.1, shall be packed in domestic type exterior containers conforming to PPP-B-591, PPP-B-636, or PPP-B-640, Class 1. Closure of containers shall be in accordance with the applicable container specification or appendix thereto. The gross weight of the packed shipping container shall not exceed the weight limitation of the applicable container specification.

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5.2.3 Level C. The pumps shall be packed in accordance with MIL-STD-1190.

5.3 Marking. All interior and exterior containers shall be marked in accordance with MIL-STD-129. Bar code marking is required and shall be in accordance with MIL-STD-129-1189.

6 NOTES

6.1 Intended use. The pump assembly is intended for use in engine oil sampling in accordance with the Army Oil Analysis Program.

6.2 Ordering data.

6.2.1 Acquisition requirements.

a. Title, number, and date of this specification.
b. First article, when required (see 4.2.1)
c. Serial number, if required (see 3.5.4)
d. Applicable levels of packing and packaging (see 5).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, maintenance manuals, drawings, parts lists, lubricating instructions, safety precautions, and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 Subject term (key word) listing.
Army Oil Analysis Program
Pump

Custodian:
Army-AL

Preparing Activity:
Army-AL
Project No:
4930-A278

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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Incl Code)* – Optional
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